Fundamental Trigonometric Identities

ClipSAT Printable Practice Suite (clipsat.org) • Topic 3.02

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

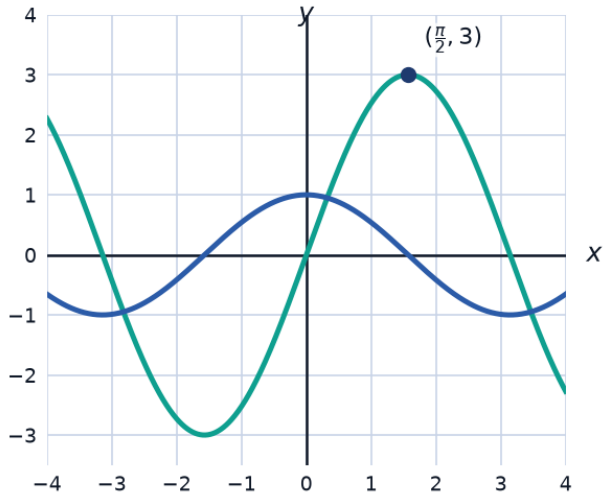
1 [Foundational] • Target: Using the Pythagorean identities

Given that $\sin\theta = \frac{3}{5}$ and θ lies in Quadrant II, find:

a $\cos\theta$ **b** $\tan\theta$ **c** $\sec\theta$ **d** $\csc\theta$

2 [Foundational] • Target: Using the Pythagorean identities

Given that $\tan\theta = 2$ and θ lies in Quadrant III, find $\sin\theta$ and $\cos\theta$ in exact simplified form.



3 [Foundational] • Target: Simplifying and verifying

Simplify each expression to a single trigonometric function:

a $\frac{1-\cos^2 x}{\sin x}$

b $\sec x \cot x$

c $\tan x \csc x$

4 [Foundational] • Target: Simplifying and verifying

Verify the identity $(1+\tan^2 x)\cos^2 x = 1$.

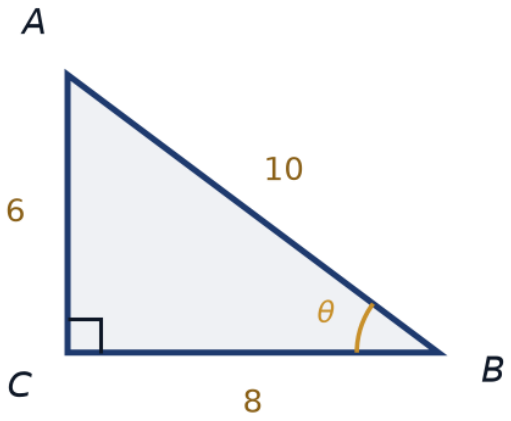
5 [Foundational] • Target: Simplifying and verifying

Verify the identity $\frac{\sin x}{1+\cos x} = \frac{1-\cos x}{\sin x}$.

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6 [Foundational] • Target: Quotient and reciprocal identities

Rewrite each expression using only $\sin x$ and $\cos x$, then simplify:



a $\cot x \sin x$

b $\frac{\sec x}{\csc x}$

c $1 - (\sin^2 x \csc^2 x)\sin^2 x$

7 [Foundational] • Target: Quotient and reciprocal identities

Simplify $\frac{\cos x}{1 - \sin x} - \tan x$ to a single term.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: More identity verifications

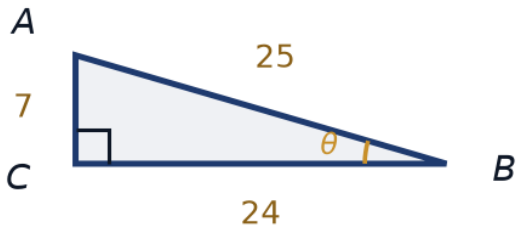
Verify the identity $\sec^2 x - \tan^2 x = 1$ using the Pythagorean identity for tangent and secant.

9 [Intermediate] • Target: More identity verifications

Verify the identity $\frac{1}{1 - \cos x} + \frac{1}{1 + \cos x} = 2\csc^2 x$.

10 [Intermediate] • Target: More identity verifications

Show that $(\sin x + \cos x)^2 = 1 + 2\sin x \cos x$.



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11 [Intermediate] • Target: More identity verifications

Given that $\cos\theta = -\frac{8}{17}$ and θ lies in Quadrant III, find $\sin\theta$, $\tan\theta$, and $\cot\theta$ in exact simplified form.

12 [Intermediate] • Target: Identities with sums and products

Verify the identity $\sin x\cos x(\tan x + \cot x) = 1$.

13 [Intermediate] • Target: Identities with sums and products

Simplify $(\sec x - \tan x)(\sec x + \tan x)$ to a constant.

14 [Intermediate] • Target: Identities involving all three Pythagorean forms

State all three forms of the Pythagorean identity (for sine/cosine, tangent/secant, and cotangent/cosecant).

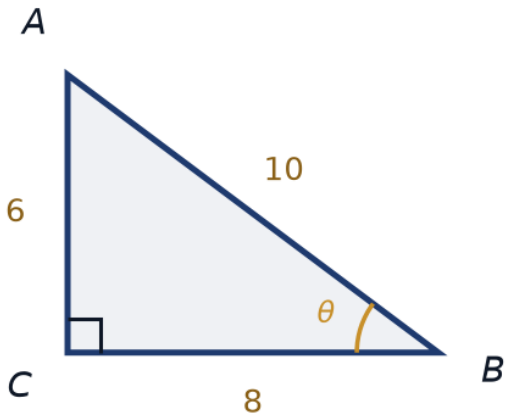
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Identities involving all three Pythagorean forms • Cross-Lesson: 8.01 Rational Functions and Asymptotes

Given $\cot\theta = \frac{5}{12}$ and θ in Quadrant I, find $\csc\theta$ and $\sin\theta$ using the cotangent-cosecant identity.

16 [Challenging] • Target: Simplifying more complex expressions • Cross-Lesson: 8.02 Graphing Rational Functions

Simplify $\frac{\sin^3 x + \cos^3 x}{\sin x + \cos x}$ using the sum-of-cubes factoring pattern.



17 [Challenging] • Target: Verifying identities with a common denominator strategy • Cross-Lesson: 8.03 Introduction to Limits

Verify $\tan x + \cot x = \sec x\csc x$.

18 [Challenging] • Target: Verifying identities with a common denominator strategy • Cross-Lesson: 9.01 Arithmetic Sequences and Series

Verify $\frac{\sin x}{\csc x} + \frac{\cos x}{\sec x} = 1$.

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19 [Challenging] • Target: Numeric identity verification (checking a specific angle) • Cross-Lesson: 9.02 Geometric Sequences and Series

Verify the identity $1 + \tan^2 x = \sec^2 x$ numerically at $x = \frac{\pi}{6}$ by computing both sides.

20 [Challenging] • Target: Identities with squared expressions • Cross-Lesson: 9.03 The Binomial Theorem

Simplify $(1 - \sin^2 x)(1 + \tan^2 x)$ to a constant.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Reciprocal and Quotient Identities

Simplify the expression $\tan x \cdot \csc x$.

a. $\sec x$

b. $\cos x$

c. $\sin x$

d. $\cot x$

22 [Foundational] • Target: Reciprocal and Quotient Identities

Simplify $\frac{\cos x}{\cot x}$.

a. $\sin x$

b. $\cos^2 x$

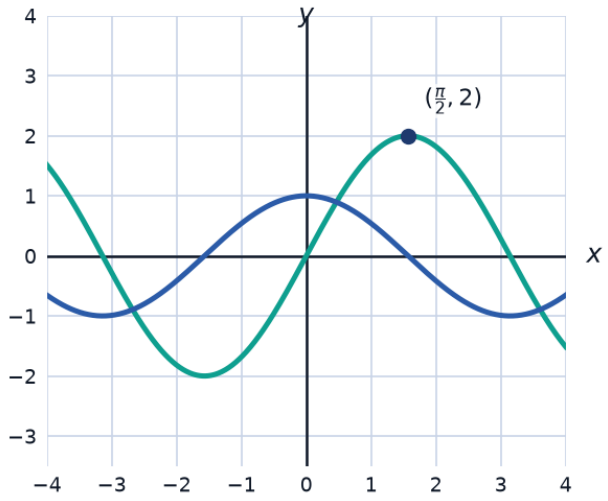
c. $\tan x$

d. $\csc x$

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23 [Foundational] • Target: Reciprocal and Quotient Identities

If $\sin\theta = -3/5$ and $\tan\theta > 0$, find $\cos\theta$.



a. $-4/5$

b. $4/5$

c. $-3/4$

d. $5/4$

24 [Foundational] • Target: Pythagorean Identities

Which identity is directly derived by dividing $\sin^2\theta + \cos^2\theta = 1$ by $\cos^2\theta$?

a. $1 + \tan^2\theta = \sec^2\theta$

b. $1 + \cot^2\theta = \csc^2\theta$

c. $\sin^2\theta = 1 - \cos^2\theta$

d. $\tan^2\theta + \cot^2\theta = 1$

25 [Foundational] • Target: Pythagorean Identities

Simplify $(\sec\theta - 1)(\sec\theta + 1)$.

a. $\tan^2\theta$

b. $\cot^2\theta$

c. $\cos^2\theta$

d. 1

26 [Foundational] • Target: Pythagorean Identities

Simplify $\frac{1 - \cos^2 x}{\sin x}$.

a. $\sin x$

b. $\cos x$

c. $\csc x$

d. 1

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27 [Foundational] • Target: Even/Odd Identities and Symmetry

Which trigonometric functions are even functions ($f(-\theta) = f(\theta)$)?

a. $\cos\theta$ and $\sec\theta$

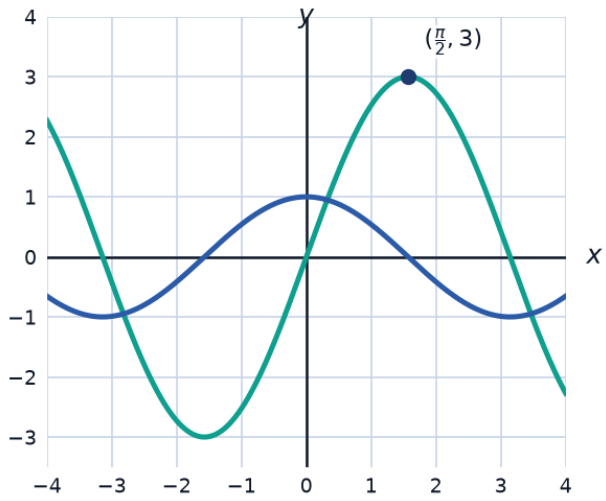
b. $\sin\theta$ and $\csc\theta$

c. $\tan\theta$ and $\cot\theta$

d. All six functions

28 [Foundational] • Target: Even/Odd Identities and Symmetry

Simplify $\sin(-\theta) \cdot \cot(-\theta)$.



a. $\cos\theta$

b. $-\cos\theta$

c. $\sin\theta$

d. $-\sin\theta$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Even/Odd Identities and Symmetry

Simplify $\frac{\tan(-x)}{\sin(-x)}$.

a. $\sec x$

b. $-\sec x$

c. $\cos x$

d. $-\cos x$

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30 [Intermediate] • Target: Verifying Identities Algebraically

Prove $\frac{\sin x}{1+\cos x} = \frac{1-\cos x}{\sin x}$. What technique verifies this algebraically?

a. Multiply numerator and denominator of the left side by the conjugate $(1-\cos x)$

b. Square both sides

c. Take the derivative

d. Cross multiply and set to 0

31 [Intermediate] • Target: Verifying Identities Algebraically

Simplify $\frac{\sec x - \cos x}{\sin x}$.

a. $\tan x$

b. $\cot x$

c. $\sec x$

d. $\sin x$

32 [Intermediate] • Target: Verifying Identities Algebraically

Simplify $(1+\tan^2\theta)(1-\sin^2\theta)$.

a. 1

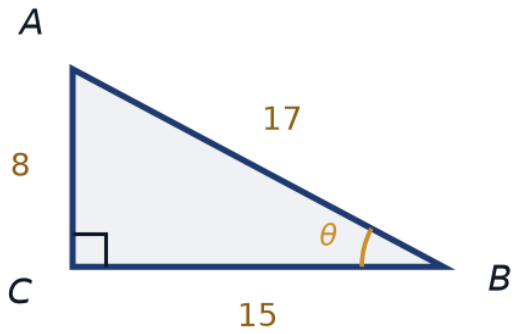
b. $\sec^2\theta$

c. $\tan^2\theta$

d. $\cos^2\theta$

33 [Intermediate] • Target: Even/Odd Identities and Symmetry

If $\cos\theta = \frac{77}{85}$ and θ is in Quadrant I, find $\sin\theta$ using the Pythagorean identity.



a. $\frac{77}{85}$

b. $\frac{36}{85}$

c. $-\frac{36}{85}$

d. $\frac{89}{170}$

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34 [Intermediate] • Target: Verifying Identities Algebraically

If $\cos\theta = \frac{80}{89}$ and θ is in Quadrant I, find $\sin\theta$ using the Pythagorean identity.

a. $\frac{80}{89}$

b. $-\frac{39}{89}$

c. $\frac{39}{89}$

d. $\frac{479}{890}$

35 [Intermediate] • Target: Using the Pythagorean identities

If $\cos\theta = \frac{40}{41}$ and θ is in Quadrant I, find $\sin\theta$ using the Pythagorean identity.

a. $\frac{131}{410}$

b. $\frac{40}{41}$

c. $\frac{9}{41}$

d. $-\frac{9}{41}$

36 [Intermediate] • Target: Simplifying and verifying

If $\cos\theta = \frac{21}{29}$ and θ is in Quadrant I, find $\sin\theta$ using the Pythagorean identity.

a. $-\frac{20}{29}$

b. $\frac{229}{290}$

c. $\frac{20}{29}$

d. $\frac{21}{29}$

37 [Intermediate] • Target: Quotient and reciprocal identities

If $\cos\theta = \frac{12}{13}$ and θ is in Quadrant I, find $\sin\theta$ using the Pythagorean identity.

a. $\frac{63}{130}$

b. $\frac{12}{13}$

c. $-\frac{5}{13}$

d. $\frac{5}{13}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: More identity verifications • Cross-Lesson: 9.01 Arithmetic Sequences and Series

If $\cos\theta = \frac{15}{17}$ and θ is in Quadrant I, find $\sin\theta$ using the Pythagorean identity.

a. $\frac{15}{17}$

b. $\frac{97}{170}$

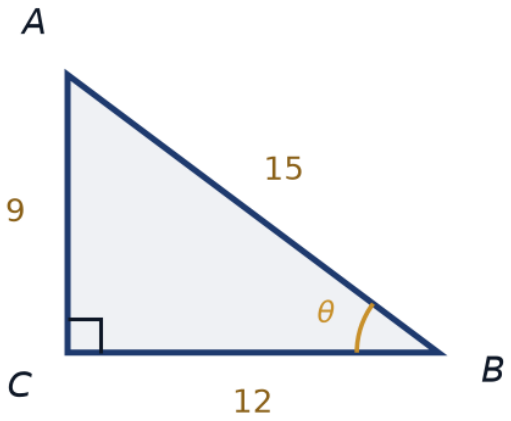
c. $\frac{8}{17}$

d. $-\frac{8}{17}$

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39 [Challenging] • Target: Identities with sums and products • Cross-Lesson: 9.02 Geometric Sequences and Series

If $\cos\theta = \frac{24}{25}$ and θ is in Quadrant I, find $\sin\theta$ using the Pythagorean identity.



a. $\frac{24}{25}$

b. $\frac{19}{50}$

c. $\frac{7}{25}$

d. $-\frac{7}{25}$

40 [Challenging] • Target: Identities involving all three Pythagorean forms • Cross-Lesson: 9.03 The Binomial Theorem

If $\cos\theta = \frac{4}{5}$ and θ is in Quadrant I, find $\sin\theta$ using the Pythagorean identity.

a. $\frac{4}{5}$

b. $-\frac{3}{5}$

c. $\frac{3}{5}$

d. $\frac{7}{10}$

41 [Challenging] • Target: Simplifying more complex expressions • Cross-Lesson: 1.01 Functions and Function Notation

If $\sin\theta = \frac{33}{65}$, find $\cos^2\theta$ using the Pythagorean identity.

a. $\frac{1089}{4225}$

b. $\frac{56}{65}$

c. $\frac{3136}{4225}$

d. $\frac{7117}{8450}$

42 [Challenging] • Target: Verifying identities with a common denominator strategy • Cross-Lesson: 1.02 Domain and Range

If $\sin\theta = \frac{48}{73}$, find $\cos^2\theta$ using the Pythagorean identity.

a. $\frac{3025}{5329}$

b. $\frac{2304}{5329}$

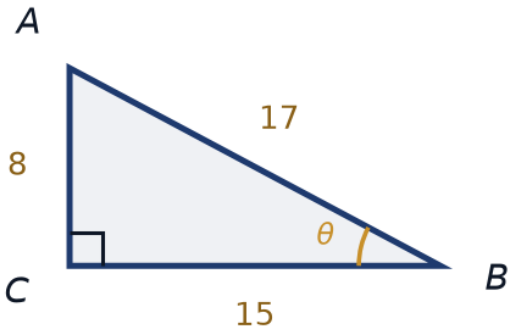
c. $\frac{55}{73}$

d. $\frac{35579}{53290}$

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43 [Challenging] • Target: Numeric identity verification (checking a specific angle) • Cross-Lesson: 1.03 Average Rate of Change

If $\sin\theta = \frac{36}{85}$, find $\cos^2\theta$ using the Pythagorean identity.



a. $\frac{5929}{7225}$

b. $\frac{1296}{7225}$

c. $\frac{77}{85}$

d. $\frac{13303}{14450}$

44 [Challenging] • Target: Identities with squared expressions • Cross-Lesson: 2.01 Right-Triangle Trigonometry

If $\sin\theta = \frac{39}{89}$, find $\cos^2\theta$ using the Pythagorean identity.

a. $\frac{1521}{7921}$

b. $\frac{80}{89}$

c. $\frac{6400}{7921}$

d. $\frac{71921}{79210}$

45 [Challenging] • Target: Reciprocal and Quotient Identities • Cross-Lesson: 2.02 Radian Measure and Arc Length

If $\sin\theta = \frac{11}{61}$, find $\cos^2\theta$ using the Pythagorean identity.

a. $\frac{121}{3721}$

b. $\frac{39721}{37210}$

c. $\frac{3600}{3721}$

d. $\frac{60}{61}$

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